LECTURE 31: MATH1061/7861
TOPICS: COUNTING SELECTIONS
DATE: TUESDAY MAY 12, 2026

Learning objectives:

(1) Reinforce techniques to solve counting problems.

(2) Learn the different ways to select r objects from n choices.

(3) Learn how to count the number of ways in each case.

Q1: Your are at an ice cream shop and can choose 3 flavours out of 10 to fill your triple scoop cone. How many different possibilities do you have?

Repetition ? Yes or No

Order matters ? Yes or No.

1) Repetion allowed + order matters.

$10 \times 10 \times 10 = 10^3 = 1000$

2) Repetition not allowed + order matters

$\frac{10!}{7!} = 10 \times 9 \times 8 = 720$

3) Repetition not allowed + order doesn't matter

$\binom{10}{3} = \frac{10!}{7!\,3!} = \frac{10 \times 9 \times 8}{3!} = 120$

4) Repetition ~~not~~ allowed + order doesn't matter

↳ Exercise: Think about.

(Not answerable).

Q2: There are 10 boys and 11 girls in a classroom. How many ways are there to choose a group of 4 students with exactly 2 girls?

# of ways of picking 2 girls:

$$\binom{11}{2} = \frac{11!}{9!\,2!} = \frac{11 \times 10}{2} = 55$$

# of ways of picking boys:

$$\binom{10}{2} = \frac{10 \times 9}{2} = 45.$$

Total: $\binom{11}{2} \times \binom{10}{2} = 55 \times 45$

$= 2475$

Q3: There are ~~30~~ 41 people in this classroom.

(a) What are the total number of possibilities for the number of birthdates (month and day)? Assume no leap years.

Repetition is allowed.

Assume order matters.

$$365^{41}$$

(b) What is the total number of possibilities ~~for~~ if no two people have ~~having~~ the same birthdates?

$$\underbrace{365 \times 364 \times \cdots \times 325}_{41 \text{ terms}} = \frac{365!}{324!}$$

(c) What is the total number of possibilities ~~for~~ if at least two people shar~~ing~~e the same birthday?

$$365^{41} - \frac{365!}{324!}$$